

Software

Get the software ready. To start with, you should have Raspberry Pi with an Internet connection. Update it to ensure that it has all the latest software and drivers. Run the command given below in the terminal to update the OS installed on Raspberry Pi.

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

Installing WiringPi library.

WiringPi is maintained under GIT, so install it with:

```
$ sudo apt-get install git-core
To obtain WiringPi using GIT:
$ git clone git://git.drogon.net/
wiringPi
```

If you are using clone operation for the first time, then use:

```
$ cd wiringPi
$ git pull origin
```

This will fetch an updated version and then run the build script using the command given below. To build/install, use the new simplified script given below:

```
$ cd wiringPi
$ ./build
```

The build script will compile and install it all for you.

Test WiringPi Installation

Run gpio command to check installation:

```
$ gpio -v
$ gpio readall
```

Setting up Apache Web server on Raspberry Pi. Apache is a popular Web server application that can be installed on Raspberry Pi to allow it to serve Web pages. On its own, Apache can serve HTML files over HTTP, and with additional modules, you can make it a dynamic Web page using scripting languages such as PHP.

Install Apache using the following:

```
$ sudo apt-get install apache2 php5
libapache2-mod-php5
```

Find Raspberry Pi IP address using the following command:

```
$ ifconfig
```

Setup

Test your server setup of Raspberry Pi from another computer by typ-



Fig. 3: Web page window

ing its IP on any Web browser on the same or different network. You should see a message saying, "It works!" This means that Apache is working.

Changing the default Web page.

The default Web page is an HTML file (index.html.) located at /var/www/html/index.html

Navigate to this directory in the terminal and have a look at what is inside.

```
$ cd /var/www/html
$ ls -al
```

This means that there is one file present inside the html directory owned by the root user. We do not need this file, so remove it.

```
$ sudo rm index.html
```

For owning the www directory, run the following command:

```
$ ls -l /var/www
```

You will see this only if you have root (the super user) privileges.

You can use the following command to change it or create a group named www-data in which you place the Pi user.

```
$ sudo chown -R pi /var/www
```

To own everything inside the directory (index.php as example), use:

```
$ sudo chown -R www-data /var/www
```

The smartfolder (inside the DVD) contains two folders, namely, html and my_project. Copy my_project inside Pi folder.

To copy html folder inside

www directory, use the following command:

```
$ sudo cp -r /home/pi/smart/html /var/www
```

Now, you have a Web server ready and it will receive the Web page after refreshing it.

We have used main.py Python script to detect interrupt and call emailscript.py. Main.py gets executed when door bell ring interrupt is received through CON3. emailscript.py Python script is used to send emails with photos as attachment.

Make necessary changes in emailscript.py. Change the text given below with your email credentials, which you will use to send emails:

```
USERNAME = "your_id@test.com"
PASSWORD = "your_password"
```

Change the email ID in sendMail (["receiver_id@test.com"]) with your email ID or the person who will be receiving the photo.

Testing

After everything (webcam, interfacing circuit and others) has been properly connected, follow the steps given below. Run main.py Python script on the terminal by giving following command:

```
$ cd my_project/
$ sudo python main.py
```

Open the browser from another computer having an Internet connection with your Raspberry Pi IP address. You will get the Web page as shown in Fig. 3. For testing purpose, provide a 3V DC at CON3 for around three seconds and release it. Press Camera tab to get the captured photo in your email. Within a few seconds you will receive a captured photo in your email attachment in receiver_id@test.com email ID. Press Lock tab to energise the sole-noid lock. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source. efymag.com



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